

NBA Player Performance Analysis Dashboard Report

Problem Statement

The NBA generates a large volume of player and team performance statistics every season. While these statistics contain valuable insights, analyzing raw data manually is challenging and time-consuming. It becomes difficult to identify top-performing players, compare team performance, evaluate shooting efficiency, and understand how factors such as age and playing position influence overall performance.

This project addresses these challenges by developing an interactive Power BI dashboard that transforms raw NBA player statistics into meaningful business insights. Through data cleaning, feature engineering, KPI tracking, and interactive visualizations, the dashboard enables users to analyze player and team performance and support data-driven decision-making.

Project Overview

The **NBA Player Performance Analysis Dashboard** is designed to analyze player and team statistics using Business Intelligence techniques. The dashboard provides interactive visualizations that help users compare player performance, evaluate team efficiency, identify scoring trends, and analyze shooting performance.

The project consists of three analytical dashboards:

- Player Performance Analysis
- Team Performance Analysis
- Shooting & Efficiency Analysis

Objective

The project aims to answer the following business questions:

Player Analysis

- Who are the top 10 players by total points?
- Which players have the highest three-point shooting percentage?
- What are the average points, assists, and rebounds by position?
- How does player age correlate with performance metrics?

Team Analysis

- Which team scored the highest total points?
- Which team has the highest field goal percentage?

- Which team has the highest rebound efficiency?
- How do teams compare in two-point and three-point shooting?

Shooting & Efficiency Analysis

- Which positions have the highest shooting efficiency?
- How does field goal percentage compare across positions?
- Which players are the most efficient shooters?
- How does age influence shooting efficiency?

Tools & Technologies

- Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Data Cleaning
- Data Visualization
- Business Intelligence

Data Preparation

Before importing the dataset into Power BI, several preprocessing and feature engineering steps were completed.

Data Cleaning

- Removed duplicate records.
- Handled missing values.
- Corrected data types.
- Verified statistical values.
- Structured the dataset for analysis.

Feature Engineering

The following calculated columns were created:

Metric	Formula
Points per Game	pts / g
Field Goal Efficiency	fg/ fga
Assist-to-Turnover Ratio	ast / tov
Rebound Efficiency	trb / mp

These calculated metrics provide deeper insights into player efficiency and overall performance.

Dataset Overview

The dataset contains statistics for:

- **478 NBA Players**
- **31 NBA Teams**

The dataset includes:

- Player Name
- Team
- Position
- Age
- Games Played
- Points
- Assists
- Rebounds
- Field Goal Percentage
- Three Point Percentage
- Two Point Percentage
- Free Throw Percentage
- Effective Field Goal Percentage

Dashboard 1 – Player Performance Analysis

Objective

Analyze player performance based on scoring, assists, rebounds, age, and position.

KPIs

- Total Players: **478**
- Total Points: **248K**
- Average Points: **519.80**
- Average Assists: **113.23**
- Average Rebounds: **219.99**

Key Insights

- The dataset includes **478 players** who collectively scored **248K points**.
- Zach Randolph, Wesley Matthews, and Wilson Chandler are among the highest scorers in the dataset.
- Stephen Curry leads in three-point shooting percentage, followed by Klay Thompson, highlighting outstanding perimeter shooting efficiency.
- Point Guards contribute the highest average assists, while Centers dominate rebounding performance.
- Players between the ages of 24 and 29 generally show stronger scoring performance than other age groups.

Dashboard 2 - Team Performance Analysis

Objective

Evaluate team performance based on scoring, shooting efficiency, and rebounding.

KPIs

- Total Teams: **31**
- Average FG%: **44%**
- Average 3P%: **24.59%**
- Rebound Efficiency: **0.18**
- Total Points: **248K**

Key Insights

- The **TOT** team records the highest total points among all teams in the dataset.
- Teams with higher rebound efficiency generally create more offensive opportunities through additional possessions.
- Two-point scoring contributes more to total team points than three-point scoring across most teams.
- Team field goal percentages remain relatively consistent, with several teams achieving efficiencies close to **50%**.

Dashboard 3 – Shooting & Efficiency Analysis

Objective

Analyze shooting performance using advanced efficiency metrics.

KPIs

- FG%: **43.63%**
- eFG%: **48.02%**
- 3P%: **24.59%**
- FT%: **69.46%**
- 2P%: **46.69%**

Key Insights

- Effective Field Goal Percentage (eFG%) provides a more comprehensive measure of shooting efficiency than traditional FG%.
- Centers achieve higher field goal percentages due to higher-percentage shot attempts near the basket.
- Stephen Curry, Klay Thompson, and Damian Lillard rank among the leading three-point shooters.
- Field Goal Percentage remains relatively stable across ages **22–32**, suggesting that shooting efficiency depends more on player role and shot selection than age alone.

Dashboard Features

The dashboard includes:

- Interactive slicers for Team, Position, Player, and Age.
- Dynamic KPI cards.
- Cross-filtering across visuals.
- Bar charts for player and team comparisons.
- Scatter plots for age versus performance analysis.
- Position-wise shooting efficiency analysis.
- Fully interactive navigation across report pages.

Overall Business Insights

- A relatively small group of elite players contributes a substantial share of the league's total scoring.
- Point Guards excel in playmaking, while Centers lead in rebounding and interior shooting efficiency.
- Players aged **24–29** demonstrate the strongest scoring performance.
- Shooting efficiency and rebounding are key contributors to overall team performance.
- Two-point scoring remains the largest contributor to team offense, although efficient three-point shooting provides a competitive advantage.
- Effective Field Goal Percentage (eFG%) offers a more accurate assessment of offensive efficiency than traditional FG%.

Recommendations

- Evaluate player performance using **eFG%** in addition to traditional FG%.
- Focus player development during the **24–29 age range**, where scoring performance is generally strongest.
- Improve rebounding efficiency to increase second-chance scoring opportunities.
- Build balanced lineups that combine strong perimeter shooters with effective rebounders.

- Use interactive BI dashboards to support player evaluation, team strategy, and performance monitoring.

Conclusion

The **NBA Player Performance Analysis Dashboard** demonstrates how Business Intelligence can transform raw sports data into actionable insights. Through data preparation, feature engineering, DAX calculations, and interactive Power BI visualizations, the project enables comprehensive analysis of player performance, team statistics, and shooting efficiency.